

Electrical Power & Safety Audit Report
AT
FM INDIA SUPPLY CHAIN PVT LTD
(Pune MCF)
RP7F+CNT, karanjvihire, Pune, Maharashtra 410501



CONDUCTED & SUBMITTED BY
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CUSTOMER DETAILS

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,
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6. Period of Audit : 16th – 18th September, 2025
7. Date of Submission of Report : 15th October, 2025

ACKNOWLEDGEMENT

We are thankful to the management of **FM India Supply Chain Pvt Ltd** for giving us this opportunity to contribute in their endeavor for efficient energy management. We are also thankful to all technical staff for co-ordination and co-operation given to us during the audit.

We also wish to place on record, the constant support and co-operation by Maintenance staff and entire technical team who were enthusiastically associated during the entire audit.

Disclaimer-

The recommendations made in the report are based on the best practices in the industry, national and international standards and good engineering judgment. NTIL does not hold responsibility for the results arising out of the implementation of the recommendations.

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- A. Summary for the past 6 month's electricity consumption. Comments on average power factor and methods to improve power factor.
- B. Exploring possibility of better use of time of use tariff Performance Evaluation of Major Utilities and Process Equipment's.
- C. To identify potentially hazardous electrical situations & provide corrective actions for same.
- D. To determine Electrical system & Electrical Safety Compliance
- E. Review and provide corrective actions for electrical safety work processes.
- F. Preparation of report based on analysis of above data
- G. Methodology & time schedule

Electrical Distribution Networks:

Power Factor: Measurements of distortion & displacement power factor at various points viz., PCC, MCC, load centers etc., and analysis of the same for optimizing the network losses.

Metering & Monitoring Status: Review of existing meters for adequate monitoring requirements.

Electrical Equipment:

Electrical measurements of all major electrical parameters viz, voltage, current, power factor, KW, frequency etc, of important equipment (motors, furnaces, electrolysis, lighting, etc) with significant energy consumption. Loads susceptible to generation of harmonics were checked for present harmonic level. System unbalance with respect to voltage and current were checked.

Lighting:

Measurements of light intensity and all major electrical parameters viz, voltage, current, PF, KW. Study of type of lamps and chokes used; day and night time lighting etc.

EXECUTIVE SUMMARY

Table -1: Inspection Summary

Sr. No.	Observation	Recommendation
1	No Surge Protection Device installed at Main Incomer Power Supply	Install 25KA 3 Phase Surge Protection device to protect the load from external spikes/surges.
2	Observed that DG power back up is not available for fire panel.	It is recommended to provide the DG power backup for fire panel.
3	Observed that all DBs wiring is found to be improper.	It is recommended to do the proper wiring in all the DB's.
4	Rubber mat not available in front of the distribution panel.	It is recommended to place a rubber mat in front of the Panel.
5	SLD not displayed in electrical room	It is recommended to display updated SLD in electrical room.
6	Observed that emergency lights are not available inside and outside the plant.	it is recommended to install emergency light at various location. Emergency DB should be separate from UPS DB.
7	Observed that dock level control wires were dropped from the top of the loading/unloading area.	Suggested to change the route of control cable.
8	Found low lux level at some places.	Install the lights of proper rating in order to have lux level as per standard.
9	All DB's are not labelled.	It is recommended to do the labelling of all DB's.
10	Found Switch board are not labeled.	It is recommended to do the proper labeling.
11	Observed that LOTO kit is not available in electrical room.	LOTO kit must be available in electrical room.
12	Found SLD on DB are not fix properly and also difficult to read due to dust	It is recommended to fix SLD properly on DB and also clean the dust
13	Found that cables are terminated without using lugs.	It is recommended to connect the cables using proper rating/size lugs

CHAPTER – 01

INTRODUCTION

1.1 Facility Brief:

Project Name	:	MCF, Pune
Facility Type	:	Warehouse
Occupancy	:	360 days
Operation Hours	:	24 Hours
Lighting Type	:	LED
Total Lighting Load	:	-
No. of Energy Meters	:	One
AC- Total Capacity Installed	:	3TR
Type of AC Units	:	Split, Duct
AC System Operation (Auto/ Manual)	:	Auto & Manual
Lighting System Operation (Auto/ Manual)	:	Manual
Dock level operation (Auto/Manual)	:	Manual
Audit Conducted on	:	16 th to 18 th September,2025

CHAPTER NO.02

ELECTRICAL LOAD ANALYSIS

2.1 Air conditioning details:

SR NO	Location	Type of indoor Unit	TR	QUANTITY	KW
1	Office Area Cabin	SPLIT	1.5	6	10.2
2		Duct	1.5	3	10.5
3	UPS Room		1.5	2	3.4
4	Severe Room		1.5	3	5.1
5		Cassette	1.5	1	1.5
TOTAL KW					30.7

General Recommendation:

- It is recommended to have servicing the air conditioning in quarterly or during the summer season by authorized vendor.

Benefits:

- No more extra cost of electricity.
- High quality of indoor air.
- No Loud Noise and breakdown of moving part.
- Good cooling ability.

2.2Lighting Details

Sr. No.	Area	Type	Qty	Watts	Total Watts
1	1 st Floor Office	2X2 Panel	30	36	1080
2	1 st Floor Office	2X2 Panel	17	36	612
3	UPS Room	2X2 Panel	4	36	144
4	Server Room	2X2 Panel	2	36	72
5	Men's Toilet	Round LED	7	18	126
6	Women's Toilet	Round LED	7	18	126
7	1 st Floor Area	Highway Round Light	5	100	500
8	Inbound Scanning Area	Highway (Round Light)	16	100	1600
9	Inbound Scanning Area	LED Lights	14	24	336
10	Docking and Transport Area	Highway (Round High)	301	100	30100
11	Dock No.5	LED Light	576	24	13824
12	Stitching Area Rack (Small Material)	LED Light (Sensor Operate)	1134	24	27216
13	High Value Area	LED Light	6	24	144
14	Reception Area	2X2 Panel	6	36	216
15	Canteen	2X2 Panel	27	36	972
Total Watt					77068
TOTAL KW					77.068

CHAPTER NO.03

POWER QUALITY ANALYSIS

Introduction

Electric power quality, or simply power quality, involves voltage, frequency, and waveform. Good power quality can be defined as a steady supply voltage that stays within the prescribed range, steady A.C. frequency close to the rated value, and smooth voltage curve waveform (resembles a sine wave). In general, it is useful to consider power quality as the compatibility between what comes out of an electric outlet and the load that is plugged into it. The term is used to describe electric power that drives an electrical load and the load's ability to function properly. Without the proper power, an electrical device (or load) may malfunction, fail prematurely or not operate at all. There are many ways in which electric power can be of poor quality and many more causes of such poor-quality power.

The quality of electrical power may be described as a set of values of parameters, such as

- Continuity of service (Whether the electrical power is subject to voltage drops or overages below or above a threshold level thereby causing blackouts or brownouts[2])
- Variation in voltage magnitude (see below)
- Transient voltages and currents
- Harmonic content in the waveforms for AC power

Equipment's used

- Fluke 435 Power Analyzer
- Insulation Tester
- ELCB Tester -1812
- Earth Tester – HTC CE8200
- Clamp Meter



Power Quality Readings:

(Main Incomer)

Sr. No.	Electrical Parameter	Unit	Quantity
1.	Total Active Power	kW	113.95
2.	Total Reactive Power	KVAr	21.143
3.	Total Apparent Power	kVA	115.130
4.	PF	-	1.00
5.	Current	Ampere	185.2/163.0/163.8
6.	Frequency	Hz	50.29
7.	Voltage THD	%	2.58
8.	Current THD %	R-Phase	11.3
		Y-Phase	9.4
		B-Phase	11.0

Findings

Observation: It is observed that Current Harmonics are more than standard acceptable limit (i.e 12%)

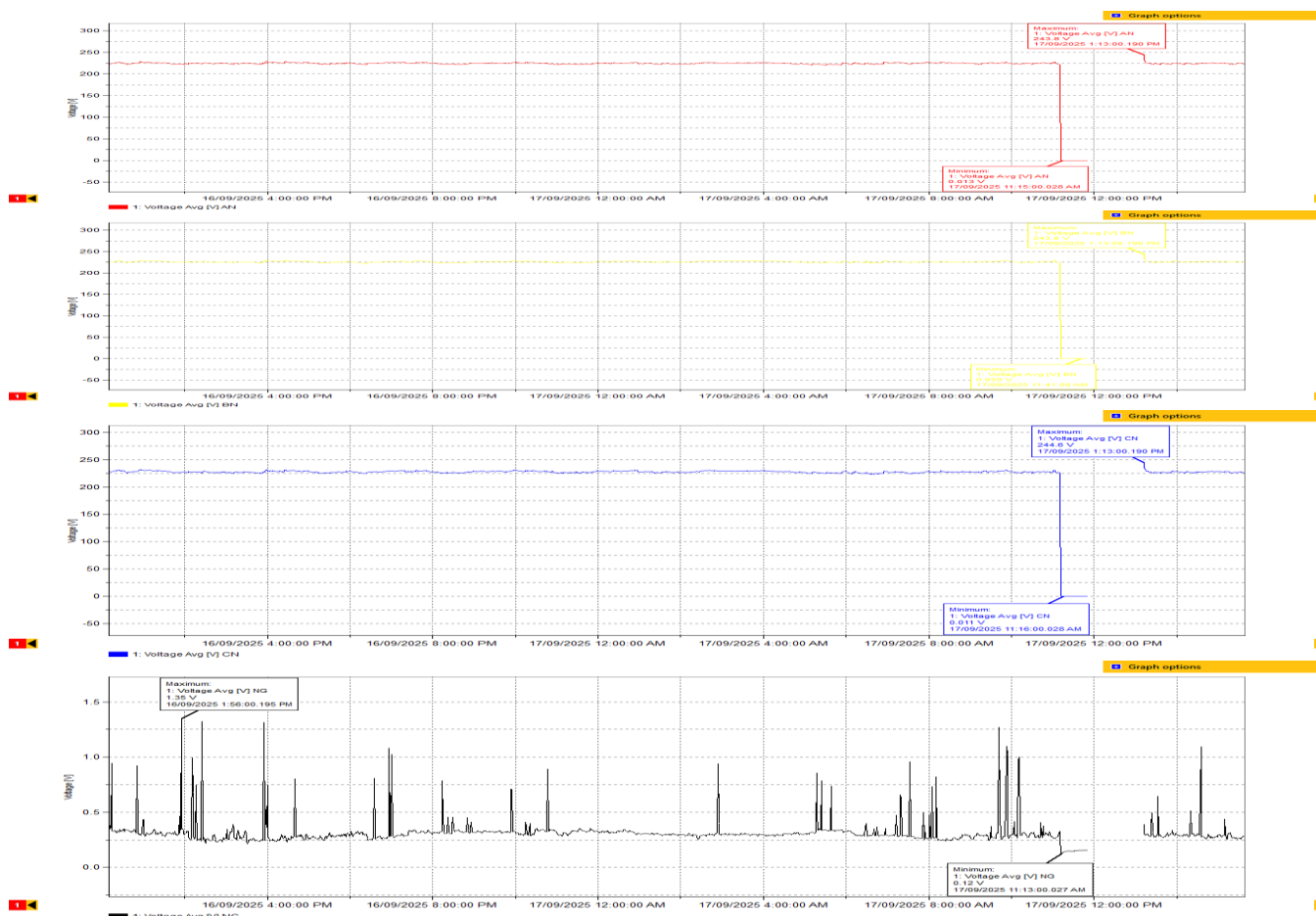
Recommendations:

-It is recommended to install 50A Active Harmonic Filter at Main Incomer to the premises which will reduce the current harmonic below standard permissible limit.

Power Quality Waveform & Trend

Location 01:1F2, 800A ACB, At Main Incomer

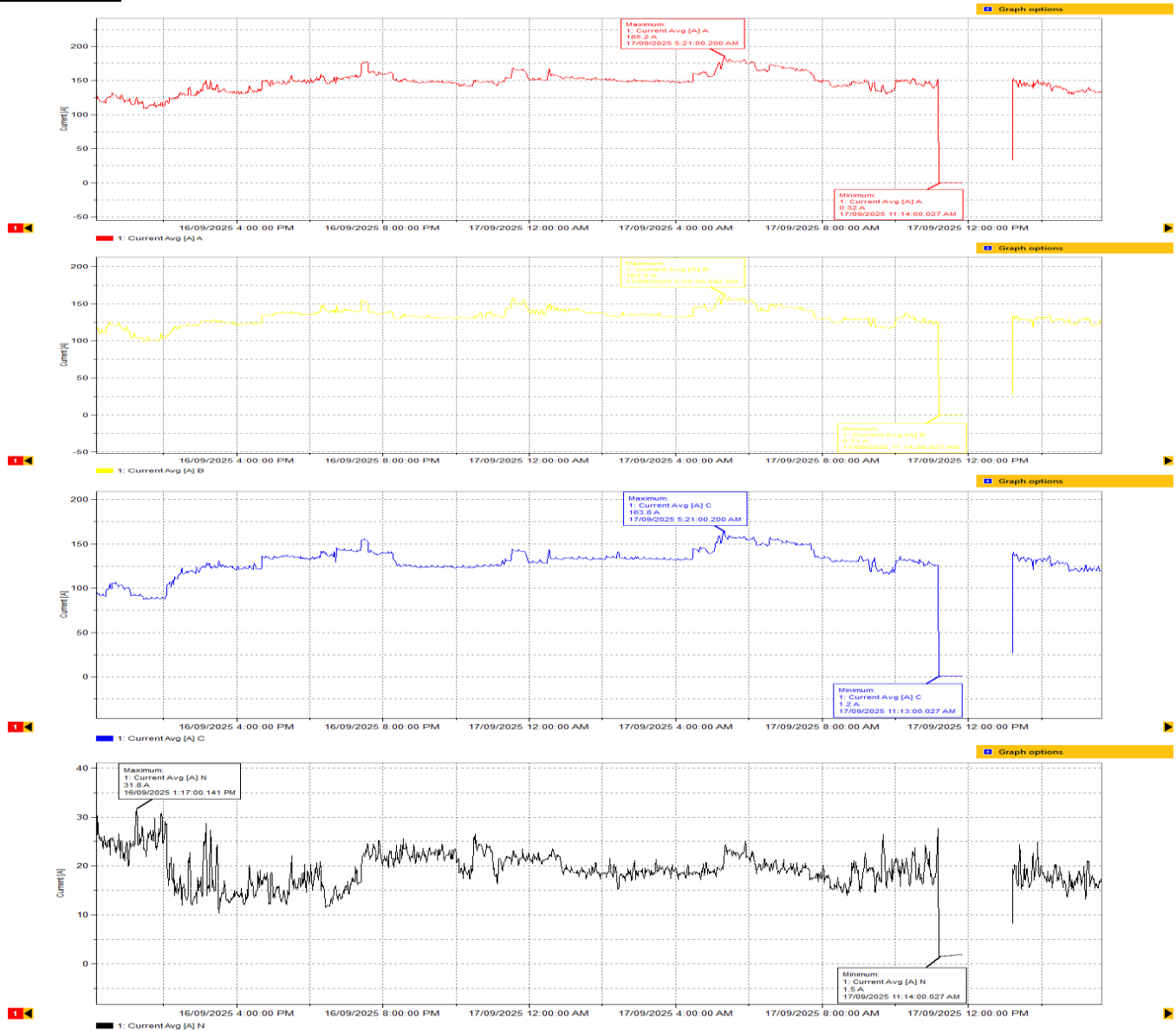
Voltage (L-N)



V rms	Min	Max
R-N	0.013	243.8
Y-N	0.059	243.8
B-N	0.011	244.6
N-E	0.12	1.35

Observation: No abnormalities found.

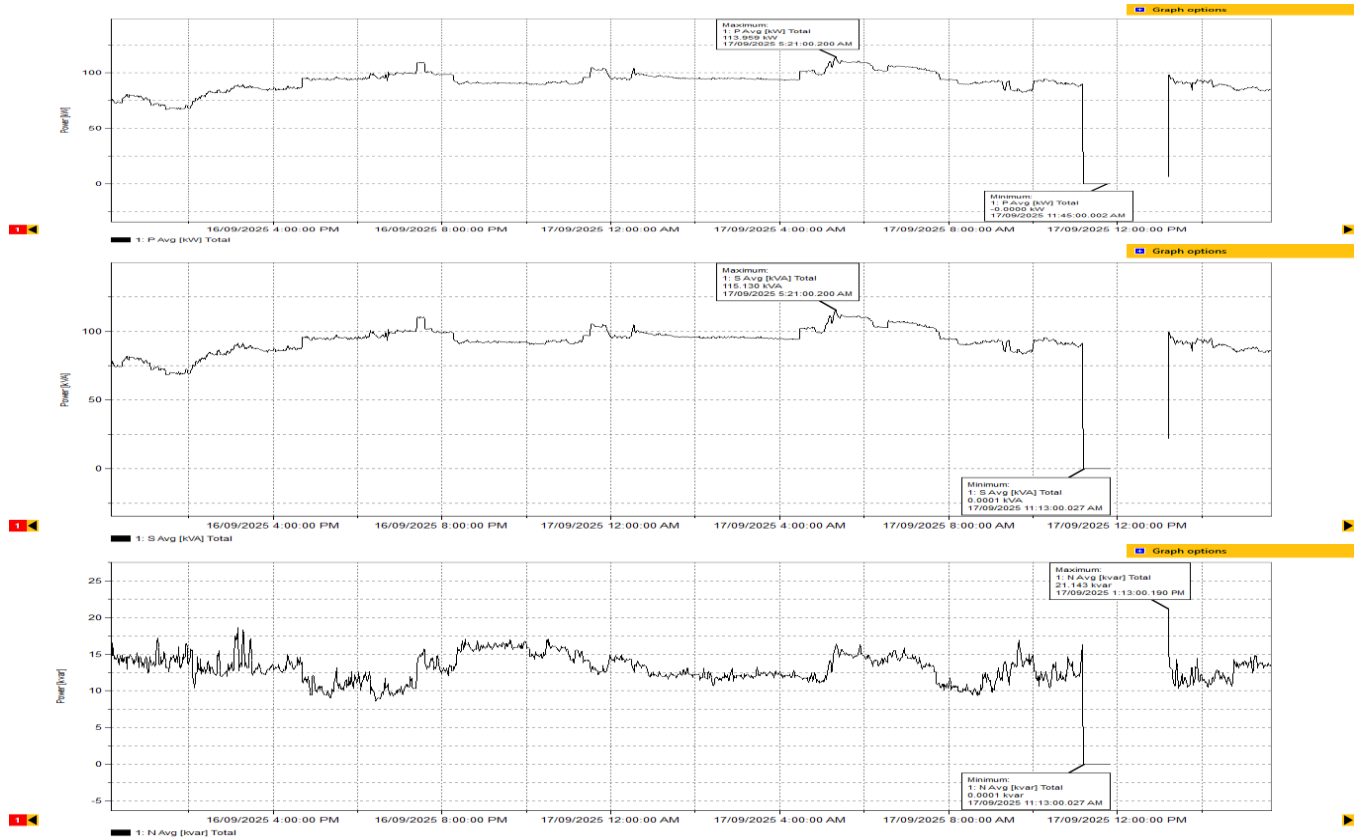
Current (A)



I rms	Min	Max
R	0.32	185.2
Y	0.31	163.0
B	1.2	163.8
N	1.5	31.8

Observation: No abnormalities found.

Power (KVA, KW, KVAR)



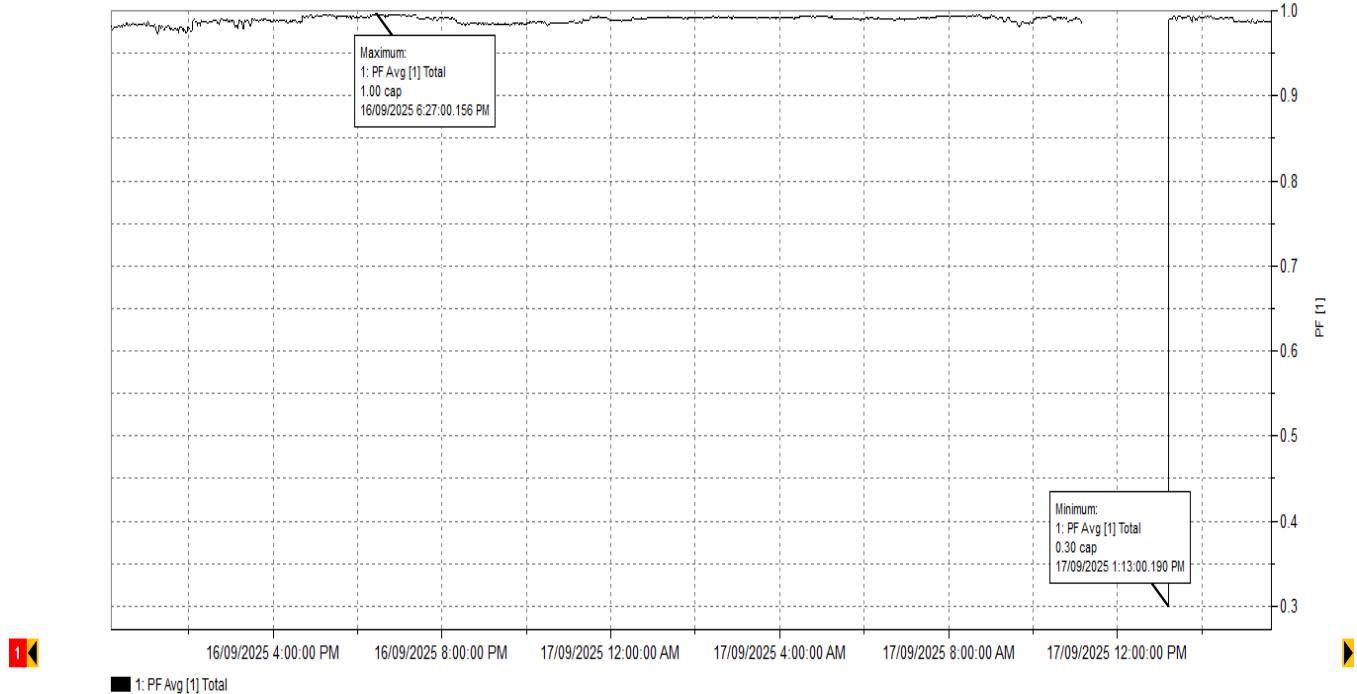
Power	Min	Max
KVA	0.0001	115.130
KW	0.00	113.95
KVAR	0.0001	21.143

Power Factor (PF)

Invalid values are not shown.

Negative PF indicates reverse power/energy.

Graph options



-	Min	Max
PF	0.30	1.00

Observation: It is observed that Power factor is maintained to unity.

Harmonic Analysis

• Harmonic Distortion Sources and Effects

Events over the last several years have focused attention on certain types of load on the electrical system that results in power quality problems for the user and utility alike. Equipment which has become common place in most facility including.

Computer power supplies

Adjustable speed drives (ASD's),

Uninterruptible power supplies (UPS's)

Are the examples of non - linear loads.

Non – linear loads generate voltage and current harmonics, which can have adverse effect of equipment's, designed for operation as linear loads (i.e. Loads designed to operate on a sinusoidal waveform of 50 or 60 Hz.)

Effects of non-linear load:

Higher heating losses in the transformers. Harmonic can have a detrimental effect on emergency generators, telephone and other sensitive electrical equipment's.

The electrical environment has absorbed most of these problems in the past. However, the problem has now reached a magnitude where Europe, the US, and other countries have proposed standards to responsible engineer system considering the electrical environment. IEEE 519-1992 and IEC 555 have evolved to become a common requirement cited when specifying equipment on newly engineered projects. The broad band harmonic filter was designed in part, to meet this specification. The present IEEE 519-1992 document establishes acceptable levels of harmonics (voltage and current) that can be introduced into the incoming feeders by commercial and industrial users. Where there may have been little cooperation previously from manufacturing to meet such specification, the adoption of IEEE 519-1992 and other similar world standards now attract the attention of everyone.

Harmonic filtering and reactive power compensation:

Various technique of improving the input currents waveform is discussed below.

- 1) Introduction of line reactors and/ or DC link chokes.
- 2) Passive filters (series, shunt, and low pass broad band filters)
- 3) Phase multiplication (12-pulse, 18-pulse rectifier systems)
- 4) Active Harmonic Compensation.

- **Harmonic Overheating:**

Harmonic distortion in power systems is generated by non-linear load equipment, and can cause substantial additional heating of the cabling system further leading to losses in energy and reducing life of the electrical system. This heating problem can be mitigated by:

- a) Installing larger size Cables.
- b) Considering the use of a harmonic filter.

- **Generation of harmonics:**

Harmonics are sinusoidal wave forms. They have frequency in multiple of fundamental frequency. They are produced by equipment using frequency conversion technology. They are also generated due to power drawing pattern of non-linear loads.

Therefore, it is essential to record power parameters and conduct harmonic study in order to provide correct solution for the harmonic filtration and power factor improvement.

- **Mitigation of Harmonic Distortion**

For resolution of harmonic problem, various techniques have been proposed to reduce the impact of load harmonic currents on the input current to an electrical system. Essentially, the intent of all these techniques is to make the input current more continuous so as to reduce the overall current harmonic distortion. The various techniques can be classified into four broad categories, viz:

- a) The use of line reactors and / or DC link chokes
- b) Installation of passive filters (series, shunt, and low pass broad band filters)
- c) Phase multiplication (12-pulse, 18-pulse rectifier system.)
- d) Harmonic compensation by active filtering.

Harmonic protection filters are used for improvement of power factor and part elimination of harmonics. They are nothing but a combination of reactors and capacitors, suitably detuned. They are useful when distortion is 15 to 40 %.

Advantages of Harmonic Filters:

- a) Protect capacitors from harmonic overloading.
- b) Eliminate 20 to 50 % predominant harmonics.
- c) Avoid harmonic amplification in the system.
- d) Improve power factor.

This system can be configured to operate in fully automatic mode. The objective is to eliminate high level of harmonic current to acceptable level to comply with utility regulation. They provide traps for individual harmonics frequency and provide reactive power compensation at fundamental frequency.

- **Harmonic filtering offers the following benefits:**

- 1 Elimination of power factor penalty.
- 2 Improvement in reduction in energy bill.
- 3 Release of additional VA capacity in the system.
- 4 Improvement in voltage profile.
- 5 Reduction in line losses
- 6 Reduction in harmonic currents.
- 7 Improvement in overall power quality.
- 8 Improvement in life of electrical equipment.
- 9 Improvement in system reliability.

IEEE 519: Recommended Practice and Requirements for Harmonic Control in Electric Power Systems

IEEE 519 is the internationally accepted standard that sets limits and guidelines for the control of harmonic distortion in electrical power systems. The most recent version is IEEE 519-2022.

Purpose of IEEE 519

To establish acceptable levels of voltage and current harmonics at the Point of Common Coupling (PCC) between the utility and the consumer, ensuring:

- Safe operation of equipment
- Compliance with utility standards
- Minimization of adverse effects on other customers or equipment

Key Definitions

- **Harmonics:** Sinusoidal components of a periodic waveform with frequencies that are integer multiples of the fundamental frequency.
- **Total Harmonic Distortion (THD):**
 - **THDv:** Voltage Total Harmonic Distortion
 - **THDi:** Current Total Harmonic Distortion
- **Point of Common Coupling (PCC):** Point where the facility connects to the utility or where multiple customers connect.

IEEE Std 519-2022 Harmonic Limits

IEEE Std 519 provides recommendations and guidelines for limiting harmonic voltage and current distortion at a point of common coupling (PCC) between the electrical system owner or operator and a user. The standard recognizes the responsibility of an electricity user to not degrade the voltage quality of the utility by drawing heavy nonlinear or distorted currents. It also recognizes the responsibility of the utility to provide users with a near sine wave voltage.

Recommended harmonic limits are shown below

Voltage Distortion Limits (At Point of Common Coupling - PCC)

Voltage Level (at PCC)	Maximum Individual Harmonic (%)	Maximum Total Harmonic Distortion (THD _v , %)
Below 1 kV	5	8
1 kV to 69 kV	3	5
69 kV to 161 kV	1.5	2.5
Above 161 kV	1	1.5

Note: These limits apply under normal operating conditions. Occasional short-duration excursions (e.g., startup) may be acceptable.

Current Distortion Limits (TDD - Total Demand Distortion)

Limits based on ratio of short-circuit current (I_{sc}) to maximum demand load current (I_L) at PCC.

I _{sc} / I _L Ratio	Maximum TDD (%)	Maximum Individual Harmonic (% of I _L)
< 20	5	4
20 – 50	8	7
50 – 100	12	10
100 – 1000	15	12
> 1000	20	15

Note:

- Values apply to odd harmonics only.
- Even harmonics should be limited to 25% of the corresponding odd harmonic limit.
- Triplen harmonics (multiples of 3) should be treated separately and filtered at the source.

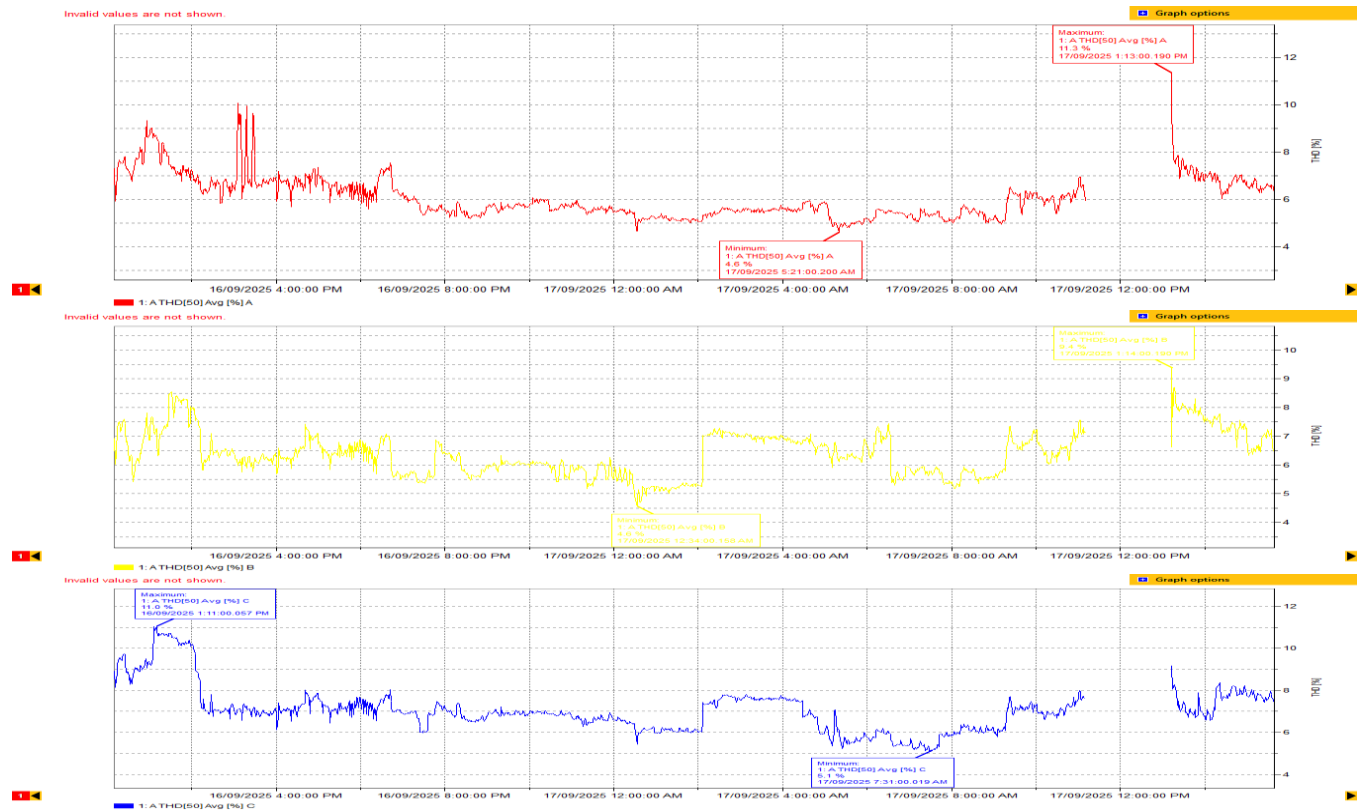
Voltage Harmonics (Main Incomer)



Minimum Voltage Harmonics	Maximum Voltage Harmonics	Maximum Acceptable Limit
0.68	2.58	8 %

It is observed that Voltage harmonics are within standard permissible limit.

Current Harmonics (Main Incomer)



Minimum Current Harmonics	Maximum Current Harmonics	Maximum Acceptable Limit
4.6	11.3	12 %

It is observed that Current harmonics are within standard permissible limit.

3.1 ELCB Test

Sr. No.	Location	ELCB/RCCB rating	Tripping current in mA			Tripping Time in ms			Remark
			R	Y	B	Rt	Yt	Bt	
1	ACDB-1 Main Incomer	63A/100mA/2P	120	100	40	19	19	13	Working Properly
2	LPDB-1	40A/100mA /2P	100	80	100	18	22	10	Working Properly
3	LDB-8	40A/100mA /4P	100	100	100	13	17	9	Working Properly
4	LDB-7	40A/100mA /2P	100	100	100	13	13	13	Working Properly
5	LDB-4	25A/100mA /4P	27	33	33	24	19	19	Working Properly
6	LDB-4	40A/100mA /4P	100	100	100	18	23	23	Working Properly

3.2 N-E Voltage Measurement

Sr. NO.	AREA	LOCATION	MEASURE LOCATION	N-E Voltage	Remark
1	OFFICE BACKSIDE	AC DB 1	MAIN I/C	1.4 V	Ok
5		LPDB 1	MAIN I/C	0.4V	Ok
9		EMDB 1	MAIN I/C	0.4V	Ok
13	RACK SIDE	ELDB-4	MCB	16.5	Weak
14		LDB-8	MAIN	3.6V	Weak
18	WAREHOUSE	LDB-7	MAIN I/C	2.9v	Ok
22		LDB-4 (EAST SIDE)	MAIN I/C	8.0v	Weak
25		SBD PANEL	MAIN I/C	0.9 v	Ok
29		LDB-4	MAIN I/C	2.4v	Ok
33		ELDB-2	MAIN I/C	7.2V	Weak
34		LPDB-4	MAIN I/C	1.7V	Ok
35	1ST FLOOR ELECTRICAL ROOM	MAIN LT PANEL	MAIN I/C -1F2	0.3V	Ok
41		MDB-1 PANEL	F1 MAIN I/C	0.3V	Ok
45	WAREHOUSE	LDB-3	MAIN I/C	3.7V	Ok
49		LDB-2	MAIN I/C	2.3V	Ok
53		ELDB-1	MAIN I/C	17.7v	Weak
54	WAREHOUSE	D SCOPE ELDB-3	MAIN I/C	2.5v	Ok
58		D SCOPE LDB-3	MAIN I/C	7.2v	Weak
61		RPDB-1	MAIN I/C	0.2v	Ok
65	WAREHOUSE	LDB	MAIN I/C	2.3v	Ok
69	SBD AREA WAREHOUSE	ELDB-3	MAIN I/C	8.6v	Weak
70		LDB-5	MAIN I/C	2.1v	Ok

74		MLDB AREA PANEL	MAIN I/C (INPUT)	2.1v	Ok
79		LDB-6	MAIN I/C	2.2v	Ok
83		LDB-9	MAIN I/C	2.2v	Ok
87	NEAR DOCK 24 WAREHOUSE SBD	LDB-02	MAIN I/C	2.9v	Ok
88	DOCK 24 AREA DSBD WAREHOUSE	ELDB-2	MAIN I/C	3.0v	Ok
89		DC DB	MAIN I/C	0.1v	Ok
90		EMDB-3	MAIN I/C	1.2v	Ok
91		LPDB-3	MAIN I/C	1.2v	Ok
92	DOCK 17 AREA	SDB-1 PANEL	MAIN I/C (CABLE TERMINATION)	0.6v	Ok
94	DOCK AREA	SERVER UPS DB-2	DB	1.0v	Ok
95	DOCK B AREA	EMDB-2	MAIN MCB	0.4v	Ok
99	DOCK 7 AREA WAREHOUSE	LPDB-2	MAIN I/C	0.1v	Ok
103	DOCK 3 AREA WAREHOUSE	LDB1	MAIN I/C	3.4v	Weak
104		ELDB-1	MAIN I/C	2.6v	Ok
105		DCDB-1	MAIN I/C	0.2v	Ok
106	DOCK 7 AREA	HULUS FAN DB	MAIN BREAKER	1.1v	Ok
107	RAMP1 AREA WAREHOUSE	LDB-10	MAIN I/C	3.1v	Weak
111	TOILET AREA ENTRY GATE	RPDB-2	MAIN I/C	0.2v	Ok
112	UPS ROOM	MDB UPS PANEL	F1	0.3V	Ok
113			F2	0.3v	Ok
115	ELECTRICAL ROOM	MAIN ELDB	MAIN I/C	2.8v	Ok

As per standard norms facility E-N Voltage should be $\leq 3V$.

Observation:

-It is observed that N-E voltage at some locations is more than standard limit.

Impact:

- In case ground failure the fault current will be get back to connected system.
- Increases rate of error or damages the equipment.
- LEDs can light up even when a small neutral voltage returns from the earth.
- Ground fault current directly has an impact on human safety.

Recommendation:

Suggested to verify the earthing connection properly connected or not.

3.3 Switch-gear analysis:

Sr. No.	Location	Measured Location	Switchgear	Current (A)				Remark
1	AC DB 1	MAIN I/C	63A 4P MCB Schneider	0.2	9	1.6	8.7	OK
5	LPDB 1	MAIN I/C	40A 4P MCB Schneider	4.6	3.1	7.1	4	OK
9	EMDB 1	MAIN I/C	40A 4P MCB Schneider	0.4	1.6	0.4	4.0	OK
13	ELDB-4	MCB	25A 2P MCB Schneider	7.7				OK
14	LDB-8	MAIN	40A 4P MCB	11.5	9.2	11.9	1.6	OK
18	LDB-7	MAIN I/C	40A 4P MCB Schneider	11.8	10.5	13.6	1.8	OK
22	LDB-4 (EAST SIDE)	MAIN I/C	25A 4P MCB Schneider	1.4	5.2		4.6	OK
25	SBD PANEL	MAIN I/C	125A 4P MCCB L&K	4.1	0.1	0.1	4.1	OK
29	LDB-4	MAIN I/C	40A 4P MCB Schneider	10.5	10.4	11.5	1.9	OK
40		5F4	400A TP MCCB	36.9	26	24	16	OK
41	MDB-1 PANEL	F1 MAIN I/C	250A 4P MCCB LK	32	26	25	11.3	OK
42		F2 O/G	125A 4P MCCB L&K	19	15.3	15.6	4.5	OK
43		F3 O/G	125A 4P MCCB L&K	4.6	0.5	0.6	4.3	OK
44		F4 O/G	125A 4P MCCB L&K	6.7	0.1	7.1	0.6	OK
45	LDB-3	MAIN I/C	40A 4P MCB Schneider	11.9	9.9	9.6	1.9	OK
49	LDB-2	MAIN I/C	40A 4P MCB Schneider	10.1	10.5	10.8	1.3	OK
54	D SCOPE ELDB-3	MAIN I/C	25A 4P 30MA RCCB Schneider	0.1	0.2	0.3	0.2	OK
58	D SCOPE LDB-3	MAIN I/C	25A 30MA 4P RCB Schneider	7.2	2		6.1	OK
61	RPDB-1	MAIN I/C	25A 30MA 4P RCB Schneider	0.1	0.4	0.1	0.3	OK
65	LDB	MAIN I/C	40A 4P MCB Schneider	11.5	10.5	10.4	1.3	OK
70	LDB-5	MAIN I/C	63A 4P MCB Schneider	8.9	10.1	10.3	1.7	OK
74	MLDB AREA PANEL	MAIN I/C (INPUT)	250A 3P MCCB L&T	98	89	92	7.5	OK
79	LDB-6	MAIN I/C	40A 4P MCB Schneider	9.5	8.7	11.1	2.6	OK
83	LDB-9	MAIN I/C	40A 4P MCB Schneider	4.9	6.6	5.3	1.6	OK

92	SDB-1 PANEL	MAIN I/C (CABLE TERMINATION)	125A 4P MCCB L&K	4.1	6.2	0.2	3.1	OK
95	EMDB-2	MAIN MCB	40A 4P MCB	1.1 2	0.1 2	0.9	0.4	OK
99	LPDB-2	MAIN I/C	40A 4P MCB Schneider	0.8	0.5	0.1	0.8	OK
105	DCDB-1	MAIN I/C	100A 3P MCCB Schneider	0.3	0.1	0.3	0.1	OK
107	LDB-10	MAIN I/C	40A 4P MCB Schneider	7.5	4.5	5.1	3	OK
111	RPDB-2	MAIN I/C	25A 4P 30MA RCB Schneider	0.1	0.2	0	0.2	OK
112	MDB UPS PANEL	F1	125A 5P MCCB	6.7	6.4	7.1	1.1	OK
113		F2	125A 3P MCCB	3.8	2.2 1	2.3	2.6	OK

Observation: No abnormalities found.

3.4 Cable analysis:

SR NO.	AREA	LOCATION	MEASURE LOCATION	N-E Voltage	Current Carrying Capacity	Cable Size	Remark
1	OFFICE BACKSIDE	AC DB 1	MAIN I/C	1.4 V	75A	4CX 16sq.mm Alu Arm	OK
5		LPDB 1	MAIN I/C	0.4V	55A	4CX10 sq.mm Alu Arm	OK
9		EMDB 1	MAIN I/C	0.4V	55A	4CX10 sq.mm Alu Arm	OK
13	RACK SIDE	ELDB-4	MCB	16.5	40A	3C X4 sq.mm Cu. Arm	OK
14		LDB-8	MAIN	3.6V	50A	4C X 6 sq.mm cu. arm	OK
18	WAREHOUSE	LDB-7	MAIN I/C	2.9v	50A	4C X 6 sq.mm cu. arm	OK
22		LDB-4 (EAST SIDE)	MAIN I/C	8.0v	32A	4C X 4sq.mm.alu.arm	OK
25		SBD PANEL	MAIN I/C	0.9 v	200A	70 sq.mm x 3.5 alu arm	OK
29		LDB-4	MAIN I/C	2.4v	50A	4C X6sq.mm cu. arm	OK
34		LPDB-4	MAIN I/C	1.7V	60A	2C X 10 sq.mm alu. arm	OK
45	WAREHOUSE	LDB-3	MAIN I/C	3.7V	50A	4C X 6 sq.mm cu. Arm	OK

49		LDB-2	MAIN I/C	2.3V	70A	4C X 10 SQMM Cu. arm	OK
53		ELDB-1	MAIN I/C	17.7v	40A	3C X4 sq.mm Cu. Arm	OK
54	WAREHOUSE	D SCOPE ELDB-3	MAIN I/C	2.5v	45A	4c x 4 sq.mm cu arm	OK
58		D SCOPE LDB-3	MAIN I/C	7.2v	45A	4c x 4 sq.mm cu arm	OK
61		RPDB-1	MAIN I/C	0.2v	70A	4c X 10sq.mm cu. arm	OK
65	WAREHOUSE	LDB	MAIN I/C	2.3v	70A	4c X 10sq.mm cu. arm	OK
69	SBD AREA WAREHOUSE	ELDB-3	MAIN I/C	8.6v	40A	3C X4 sq.mm Cu. Arm	OK
74		MLDB AREA PANEL	MAIN I/C (INPUT)	2.1v	100A	18 sq. mmx 3.5c alu arm	OK
79		LDB-6	MAIN I/C	2.2v	70A	4C X 10 SQMM Cu. arm	OK
83		LDB-9	MAIN I/C	2.2v	50A	4C X 6 sq.mm cu. arm	OK
88	DOCK 24 AREA DSBD WAREHOUSE	ELDB-2	MAIN I/C	3.0v	40A	3C X4 sq.mm Cu. Arm	OK
90		EMDB-3	MAIN I/C	1.2v	60A	2c x 10 sq.mm alu arm	OK
91		LPDB-3	MAIN I/C	1.2v	60A	2c x10sq.mm alu arm	OK
94	DOCK AREA	SERVER UPS DB-2	DB	1.0v	60A	2C X 10 sq.mm alu. arm	OK
95	DOCK B AREA	EMDB-2	MAIN MCB	0.4v	55A	4cx10sq.mm alu arm	OK
99	DOCK 7 AREA WAREHOUSE	LPDB-2	MAIN I/C	0.1v	75A	16sq.mm x 4c alu. arm	OK
103	DOCK 3 AREA WAREHOUSE	LDB1	MAIN I/C	3.4v	40A	3cx4sqmm.cu. arm	OK
104		ELDB-1	MAIN I/C	2.6v	40A	3cx4sqmm.cu. arm	OK
106	DOCK 7 AREA	HULUS FAN DB	MAIN BREAKER	1.1v	60A	2C X 10 sq.mm alu. arm	OK
107	RAMP1 AREA WAREHOUSE	LDB-10	MAIN I/C	3.1v	70A	4cx 10sq.mm cu arm	OK
111	TOILET AREA ENTRY GATE	RPDB-2	MAIN I/C	0.2v	70A	4cx 10sq.mm cu arm	OK
115	ELECTRICAL ROOM	MAIN ELDB	MAIN I/C	2.8v	70A	4cx 10sq.mm cu arm	OK

Observation:

- All cable are under standard current carrying capacity

3.5 Lux Level Measurement

Sr. No.	Area	Location	Lux Value	Standard Lux Level	Remark
1	warehouse	Reception	256	150	Normal
2			232	150	Normal
3		Gr. Floor Pantry	276	150	Normal
4			259	150	Normal
5			243	150	Normal
6		1ST FLOOR OFFICE	876	150	Normal
7			889	150	Normal
8			799	150	Normal
9			821	150	Normal
10			873	150	Normal
11		Reception passage	233	150	Normal
12			289	150	Normal
13		stairs step 26	56	150	Alert
14		stairs step 10	28	150	Alert
15		1st floor Pantry	567	150	Normal
16			489	150	Normal
17			523	150	Normal
18		Lt Room Front	211	150	Normal
19			243	150	Normal
20			167	150	Normal
21		Lt Room panel back	210	150	Normal
22			999	150	Normal
23	Warehouse Outside	Fire Exit door 1	976	150	Normal
24		Fire Exit door 2	890	150	Normal
25		Fire Exit door 3	33	150	Alert
26		Fire Exit door 4	46	150	Alert
27		Fire Exit door 5	132	150	Alert
28		Fire Exit door 6	5	150	Alert
29		Fire Exit door 7	7	150	Alert
30		Fire Exit door 8	193	150	Normal
31		Fire Exit door 9	551	150	Normal
32		Fire Exit door 10	273	150	Normal
33		Fire Exit door 11	581	150	Normal

34		Fire Exit door 12	122	150	Normal
35		Fire Exit door 13	5	150	Alert
36		Fire Exit door 14	9	150	Alert
37		Fire Exit door 15	73	150	Alert
38		Fire Exit door 16	143	150	Alert
39		Fire Exit door 17	19	150	Alert
40		Fire Exit door 18	315	150	Normal
41		Fire Exit door 19	25	150	Alert
42		Fire Exit door 20	3	150	Alert
43	RACK AREA	BETWEEN 2-3	9	150	Alert
44			115	150	Alert
45			117	150	Alert
46			53	150	Alert
47			9	150	Alert
48			1	150	Alert
49			4	150	Alert
50			170	150	Normal
51		BETWEEN 5-6	181	150	Normal
52			201	150	Normal
53			192	150	Normal
54			220	150	Normal
55			168	150	Normal
56			245	150	Normal
57			161	150	Normal
58		BETWEEN 13-14	120	150	Alert
59			253	150	Normal
60			270	150	Normal
61			250	150	Normal
62			268	150	Normal
63			236	150	Normal
64			259	150	Normal
65			186	150	Normal
66			220	150	Normal
67		BETWEEN 21-22	240	150	Normal
68			251	150	Normal
69			276	150	Normal
70			307	150	Normal
71			328	150	Normal
72			278	150	Normal
73			178	150	Normal
74	SBD AREA RACK	BETWEEN 15-16	174	150	Normal

75			298	150	Normal
76			263	150	Normal
77			217	150	Normal
78			232	150	Normal
79			219	150	Normal
80			203	150	Normal
81		BETWEEN 25-26	262	150	Normal
82			219	150	Normal
83			209	150	Normal
84			202	150	Normal
85			255	150	Normal
86			233	150	Normal
87			128	150	Alert
88			128	150	Alert
89		BETWEEN 29-30	224	150	Normal
90			259	150	Normal
91			226	150	Normal
92			212	150	Normal
93			206	150	Normal
94			223	150	Normal
95			393	150	Normal
96			363	150	Normal
97			239	150	Normal
98			228	150	Normal
99			198	150	Normal
100			239	150	Normal
101			241	150	Normal
102			219	150	Normal
103			157	150	Normal

Observation:

- Found some faulty or damaged lights in plant.
- Observed that most of the location has low lux level.

Impact:

- Low lux level increases accidental damaged of productivity.
- It will be affected on efficiency of work.
- Workers are unsafe when operating forklift & dock area.

Recommendation:

-The Faulty or damaged light should be replaced as same rating.

3.7 Earth pit Testing:

Sr. No.	AREA	EARTH PIT NO.	Measured Value (Ω)	Remark
1	TRANSFORMER AREA	EARTH PIT NO-1	2.7	Ok
2		EARTH PIT NO-2	3.5	Ok
3		EARTH PIT NO-3	0.94	Ok
4		EARTH PIT NO-4	0.48	Ok
5		EARTH PIT NO-5	0.45	Ok
6		EARTH PIT NO-6	0.57	Ok
7		EARTH PIT NO-7	0.5	Ok
8		EARTH PIT NO-8	0.65	Ok
9	SECURITY CABIN SIDE	EARTH PIT NO-9	0.85	Ok
10		EARTH PIT NO-10	0.64	Ok
11		EARTH PIT NO-11	0.51	Ok
12		EARTH PIT NO-12	0.56	Ok
13		EARTH PIT NO-13	0.86	Ok
14		EARTH PIT NO-14	0.53	Ok
15	TRANSFORMER FRONT AREA	EARTH PIT NO-15	0.52	Ok
16		EARTH PIT NO-16	4.6	Ok
17		EARTH PIT NO-17	4.5	Ok
18		EARTH PIT NO-18	3.06	Ok
19		LT PANEL EP NO-1	0.38	Ok
20		LT PANEL EP NO-2	0.51	Ok
21		LIGHTNING EP. 1	0.9	Ok
22		LIGHTNING EP. 2	0.83	Ok
23		LIGHTNING EP. 3	0.51	Ok
24	DG AREA	DG BODY E.P 1	0.52	Ok
25		DG BODY E.P 2	0.86	Ok
26		DG NEUTRAL E.P 1	0.94	Ok
27		DG NEUTRAL E.P 2	2.1	Ok

As per standard norms earth pit resistance should be ≤ 5 ohm

Observation:

-It is observed that earth pit resistance values of all the earth pits are within standard limit.

CHAPTER NO.05

ONSITE OBSERVATIONS IMPACTS, STANDARDS AND RECOMMENDATIONS

Audit findings are giving risk ratings to understand the risk level involved and take necessary steps to avoid any kind of incidents or accidents. Risk ratings for this audit are defined as below and observations are given in normal font whereas recommendations are given in italic bold font.

RISK RATING OF AUDIT FINDINGS	
Rating 1 (R1) High Risk (Immediate Action Required) Observation: RED Colour	Situations, which could result in substantial risk to the safety of the employees, environment, stockholders, customers, the company or in criminal or civil liability for knowing violations. Some administrative issues could be categorizes as level 1. (e.g. failure a report a significant release when required)
Rating 2 (R2) Medium Risk (Monitor & Take Action As Earliest Convenient) Observation: Yellow	Does not meet the criteria for level 1 but is more than an isolated occasional situation. Should not continue beyond the short term. This category can usually result in a notice of violation from regulatory agencies. It could also be serious void and/ or breakdown in management system.
Rating 3 (R3) (No Action Required) Observation: Green	Findings may be administrative nature or involve an isolated occasional situation. They may involve temporary or occasional instances of non-compliance. They may also involve opportunities for substantive system improvement.

Sr. No.	Observations	Impacts/Condition	Reference Standards & Recommendation	Risk Ratings
Electrical Room				
1.	Authorized person list should be prepared and displayed in electrical room: List is not displayed at any electrical room	Available		
2.	Checking for rubber matt in front of panel	Not Available	As per INSULATING MATS FOR ELECTRICAL PURPOSES IS 15652:2006 proper rubber mat having proper insulation level is need to place near panels	R2
3.	First Aid box in electrical rooms	No First Aid Box is available in the room.	As per IS 5216 practice The first Aid box is kept with primary medicine	R2
4.	Checking any door is damaged or not closed properly	Not found	As per IS 5216 all panel/ MCB/ D.Bs door are properly closed, the cable laying properly on the trench and all cable start and end shall be marked with metal labels to indicate the cable origin and destination.	R2
5.	Checking of general cable conditions and its laying method	Not done properly		
6.	Checking for cable Marking for area and Switch Gears	Not available		
7.	Power Factor	Maintained	-	
8.	Checking for installation of ELCB/RCCB at final distribution	Available	As per IS 5216 standard practises installed ELCB for all power loads and lighting loads	
9.	Is there any SLD of electrical installation	Yes, available, but not at main panel room	Updated single line and the electrical layout drawing should be prepared and displayed at DB, but require to display at Main Panel room	R2

10.	Earthing is most important part in electrical installation. Earthpit no's are not clearly identified. The earthpit resistance value is not mentioned. All earth pits need to cleaning and maintenance is required.	If earthing is not proper, chances of electrical shock to the human beings & also to chance of fire hazards	As per IS 3043 regular maintenances of each earth pit shall be carried out. Earth pit resistance value can be measured and maintained accordingly and record should be maintained.	R1
11.	All Earth pits are not traceable. No no's marked on the earth pits. The resistance value is not mention on the earth pits.	Regularly checking of the earth pits and mention the resistance value on the Earth pits help to identified the condition of earth pits	As per IS 3043: 1987 the earth chamber should be properly covers to maintained unwanted soil inside the chambers as well as mention the resistance value on the earth pits	R2
12.	The main panel condition in terms of maintenance:	The panel is in good condition. But behind of the main breaker protection cover not available.	All live part of electricity should cover by acrylic/Bakelite sheet.	R1
13.	Main Panel: Is there available Opening hole	No	As per IS standard recommendation need to closed empty entry by dummy plate to protect any human shocked as well protects small animal entry.	-
14.	There is no provision of emergency Tripping of Mains from Remote area	No; Emergency tripping is required to immediate action require to avoid major accident in the electrical equipment	As per the IS 5216: 1982 need to installed emergency tripping help to take immediate action to avoid major accident in the electrical equipment.	R3
15.	Emergency light availability	Available		
16.	Termination condition in all panels	Available		

17.	The outgoing cable joints & cable lying found load tension due to improper lying of the cable.	This will impact on the cable bending. Its occurs short circuiting, cable heating	As per IS 5261 electrical proper cable support is require in distribution.	R3
18.	Danger boards on Panel	Available		
19.	Shock treatment chart at panel room	There is no shock treatment chart in the panel room.	Shock treatment chart with two Languages to be displayed.	R2
20.	Layout drawings of earthing Pits and its route of Earth strip is not prepared.	For easy identification and immediate action in emergency	Layout drawing of the Earthing strips its route and the Respective Pit should be marked and Labelled.	R2
21.	Some areas found that the terminations of Cables are not done properly by using Gland and Lugs. D.B Cover is not closed. There are openings left in D.Bs	-	All such things are need to rectify	R2
22.	Exhaust fan is not available in electrical room.	If room temperature will not be maintained. This will reduce the efficiency of equipment.	As per the standard exhaust fan should be install in electrical room	R2
23.	Lock out tag out kit not available at electrical room.	Worker's servicing or maintaining machines or equipment may be seriously injured or killed if hazardous energy is not properly controlled.	As per electrical standard LOTO kit should be available in electrical room.	R1
24.	DG power backup for fire panel.	Available		

25.	Emergency power point is available at DG breaker.	-	As per electrical standard need to arrange power point at DG location	R2
26.	Outdoor MCB is not enclosed with waterproof enclosure.	Short circuit may occur during rainy season.	As per electrical standard outdoor electrical breaker should be place in waterproof enclosure.	R1

GENERAL RECOMMENDATIONS

As per Regulation 12 by CEA (Measures relating to Safety and Electric Supply)

1. Distinct and prominent naming of the equipment's building/control room shall be ensured.
2. Single Line Diagram to be displayed in the electrical control rooms.

Switchgear room:

- Annual maintenance work must be carried out by company itself. It should be sure that only trained person with valid license are allowed for electrical work.
- Display emergency phone numbers (Ambulance, Police, and Fire Bridge) at all working areas.
- Check before handling electrical equipment's technicians should wear all PPE's.
- Proper identification of panels along with details of incoming/outgoing shall be done.
- Incoming and outgoing shall be named on the panels.
- Do not store unwanted scraps behind Panels.
- Door opening interlock for all the panels shall be ensured before energization.
- Unused spare holes in the panels shall be plugged.
- All the panels shall be double earthed
- Warning notice "Isolate Supply before opening" shall be written on the panels.
- Panel locking shall be ensured & Missing bolts, if any, shall be provided

REFERENCES

1. Insulating mats for electrical purposes is 15652:2006
2. IS 3043: 1987 for Earthing
3. Asha - PPES standards For Electrical Equipment's. IS 5572: 1994
4. Indian Electricity Rule 1956
5. CEA Regulations 2010
6. IS5216 - Recommendation on safety procedure and practices in electrical Work.
7. IS 5216 : 1982 Part I Recommendations on safety procedures and practices in electrical work-
General
8. IS 3043: Code of Practice for Earthing IS 818: 1968 Code of practice for Safety and Health
Requirements in electric and gas welding and cutting operations.

Location: - Warehouse		Warehouse  Warehouse Toilet Area  Warehouse LDB-2  Warehouse LDB-4  UPS Room
Risk Category: - R2		
Finding: - Found that Insulation mat is not placed near Electrical Panels/DB's.		
Recommendation: - It is recommended to place the Insulation mat near all Electrical Panels/DB's		

Location: - MDB UPS Panel	
Risk Category: - R2	
Finding: - Found that cable inserted without using glands .	
Recommendation: - It is recommended to use cable glands of proper size during entry or exit of cables from panels/DB	

Location: - Main LT Panel- Main Incomer	
Risk Category: - R1	
Finding: - Found that cables are terminated without using lugs.	
Recommendation: - It is recommended to connect the cables using proper rating/size lugs	

Location: - Dock7 Area HVLS Fan DB	
Risk Category: - R1	
Finding: - Found body earthing not provided to DB	
Recommendation: - It is recommended to provide earthing to electrical DB	

Location: - Dock-7 Area	
Risk Category: -R2	
Finding: - Found 2Run, 2C X10Sq.mm cable used for 3Phase supply instead of 4C cable.	
Recommendation: - It is recommended to use 4C X10Sq.mm cable for 3Phase cable.	

Location: - Dock-3 Area	
Risk Category: -R1	
Finding: - Found Phase & Neutral sequence interchanged in MCB .	
Recommendation: - It is recommended to correct the wiring by properly connecting the Phase and Neutral as per standard polarity	

CHAPTER NO.06

THERMOGRAPHY

Infrared Thermal Imaging-Principle:

All objects that have a temperature above absolute zero (0 Degree Kelvin) emit infrared radiation, which can be measured on the infrared spectral band of the electromagnetic spectrum. The technique of measuring & viewing this energy is called infrared thermography or thermal imaging. Most thermal imaging systems respond to wavelengths of 3 – 5 micrometers or 8 –12 micrometers.

Why Thermal Imaging?

Thermography is nothing but ‘capturing heat images on film’. Swedish power board began inspecting a large number of electrical components (1.5 lakhs plus) way back in 1965 & UK power generation board began effectively utilizing Thermography for predictive maintenance of transmission lines.

The application of both Thermal imaging & Ultra sonic detection techniques are based on a fundamental principle, ‘Electrical Failures are preceded by rise in temperature & sound’. If you understand this basic principle, carry out effective techniques to detect these failure ‘precursors’ and take timely corrective action, electrical equipment failures can be avoided.

Although T-imaging applications are numerous (quality checks, tank level monitoring, detection of steam leaks, integrity of furnace refractory lining, etc.), we will discuss electrical applications in this application note.

The electrical maintenance practices evolved over a period of time. Breakdown to preventive to predictive to risk /reliability based maintenance practices. Logically, high reliability plant & equipment requires to be maintained accordingly. In countries like USA, UK, France, Netherlands, etc., infrared Inspections are now recognized and recommended as a best practice for the inspection of electrical installations. In order to assess the equipment condition by predicting failures, without losing production, applications such as thermal imaging are gaining importance.

Disclaimer:

The recommendations made in the report are based on the best practices in the industry, national and international standards and good engineering judgment. NTIL does not hold responsibility for the results arising out of the implementation of the recommendations.

Total location monitored	118
Risk Level	Number
Critical	0
Severe	1
Alert	3
Normal	114

1.2 The Thermo graphic Electrical Inspection

The intent of the Thermo graphic Electrical Inspection is to assist in detecting potential equipment failure by measuring abnormal temperature signatures of Electrical Distribution System Components. By evaluating the operating condition of these components, potential failures can be located and problem severity determined as part of a structured ongoing maintenance program. This will provide your organization with a predictive and proactive risk management program where outages and repairs of your electrical system can be avoided.

1.3 Background

All objects around us including ourselves are constantly emitting thermal radiation (heat). This technique for making energy visible is called Infrared Thermography. The thermal image captured by the thermo graphic camera allows for meaningful interpretation. Data it contains is digitally stored and computer processed for further analysis. This information is stored for future reference and trend analysis.

1.4 Equipment Used

Make: Fluke

Model: Ti32



1.5 Report Format

Any thermal anomalies are presented on a report page as a thermal image and a digital photograph of the electrical component. To best determine the severity of a problem direct temperature measurement of the equipment and the temperature differential of the faulty components are provided.

1.6 Temperature Differential

Areas of electrical components operating at higher temperatures than that of adjacent areas, comparisons to ambient air temperatures, and enclosures of electrical apparatus are typically an indication of a problem with the higher temperature part.

For every problem, components are always inspected for physical damage to determine if the component should be replaced rather than repaired. An inspection should be made after a problem has been repaired to ensure it has been properly corrected.

As a guide to determining the severity of a problem we recommend for any of the three classifications of infrared testing on electrical components. It should be noted that these temperature criteria are employed in conjunction with other analysis procedures and tools in the interest of accurately identifying problem areas:

Analysis by using Marker on Images:-

SIMILAR COMPARISONS - Components/Conductors that have equal or near equal loading

Problem level	Temperature Differential	Description
Normal	<5°C	No action required
Alert	5°C - 8°C	First Stage. Monitor and repair during next scheduled maintenance
Severe	8°C - 15°C	Second Stage. Should be repaired before scheduled maintenance.
Critical	>15°C	Acute Stage. Should be repaired immediately
References: NFPA - 70B Standard, NETA, and EPRI		

Analysis by using Rectangle/Square on Images:-

AMBIENT COMPARISONS - Components/Conductors with no other items of similar load and ambient air is used as a reference

Problem level	Temperature Differential	Description
Normal	<10°C	No action required
Alert	10.1°C - 20°C	First Stage. Monitor and repair during next scheduled maintenance
Severe	20.1°C - 40°C	Second Stage. Should be repaired before scheduled maintenance.
Critical	>40°C	
References: NFPA - 70B Standard, NETA, and EPRI		

INDIRECT COMPARISONS - Temperature observed across enclosed, electrical compartments and Liquid/oil filled electrical items

Problem level	Temperature Differential	Description
Normal	<1°C	No action required
Alert	1.1°C - 3°C	First Stage. Monitor and repair during next scheduled maintenance
Severe	3.1°C - 5°C	Second Stage. Should be repaired before scheduled maintenance.
Critical	>5°C	Acute Stage. Should be repaired immediately
References: NFPA - 70B Standard, NETA, and EPRI		

The above criteria provide standardized guidelines when determining if a thermal anomaly warrants further investigation based on temperature difference. Other factors such as ambient temperature, environmental conditions, criticality of the equipment, and electrical analysis tests must be considered.

Standards For Component Analysis: -

Sr. No.	Device	Standard	Terminal Temp Rise	Max Surface Temp(Typical)
1	MCB	IS/IEC 60898	45-55°C	~85°C
2	MCCB	IS/IEC 60947-2	70°C	~100°C
3	Contactor	IS/IEC 60947-4-1	70°C(term), 75°C (coil)	~105°C
4	ACB	IS/IEC 60947-2	70°C	~100°C

Thermography summary sheet-

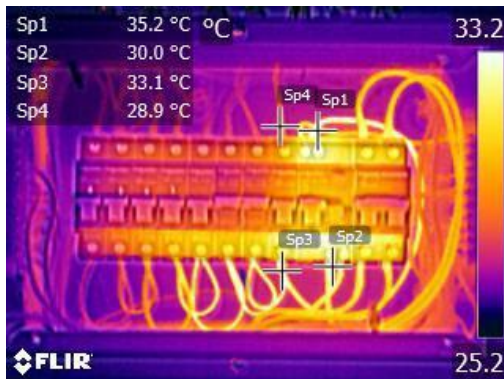
SR NO.	AREA	LOCATION	MEASURE LOCATION	IMAGE NO.	RISK CATEGORY
1	OFFICE BACKSIDE	AC DB 1	MAIN I/C	6682	Normal
2			R PHASE	6683	Normal
3			Y PHASE	6684	Normal
4			B PHASE	6685	Normal
5		LPDB 1	MAIN I/C	6686	Normal
6			R PHASE	6687	Normal
7			Y PHASE	6688	Normal
8			B PHASE	6689	Normal
9		EMDB 1	MAIN I/C	6690	Normal
10			R PHASE	6691	Normal
11			Y PHASE	6692	Normal
12			B PHASE	6693	Normal
13	RACK SIDE	ELDB-4	MCB	6694	Alert
14		LDB-8	MAIN	6695	Normal
15			R PHASE	6696	Normal
16			Y PHASE	6697	Normal
17			B PHASE	6698	Normal
18	WAREHOUSE	LDB-7	MAIN I/C	6699	Normal
19			R PHASE	6700	Normal
20			Y PHASE	6701	Normal
21			B PHASE	6702	Normal
22		LDB-4 (EAST SIDE)	MAIN I/C	6703	Normal
23			R PHASE	6704	Normal
24			Y PHASE	6705	Normal
25		SBD PANEL	MAIN I/C	6706	Normal
26			O/G MCB FEEDER 1	6708	Normal
27			O/G MCB FEEDER 2	6709	Normal
28			O/G MCB FEEDER 3	6707	Normal
29		LDB-4	MAIN I/C	6710	Normal
30			R PHASE	6711	Normal
31			Y PHASE	6712	Normal
32			B PHASE	6713	Normal
33		ELDB-2	MAIN I/C	6714	Normal
34		LPDB-4	MAIN I/C	6715	Normal
35	1ST FLOOR ELECTRICAL ROOM	MAIN LT PANEL	MAIN I/C -1F2	6716	Normal
36			MLDB O/G -4F2	6717	Normal
37			5F1-O/G	6719	Normal

38			5F2-DCDB-1 O/G	6720	Normal
39			5F3-DCDB-2	6721	Normal
40			4F4	6718	Normal
41		MDB-1 PANEL	F1 MAIN I/C	6722	Normal
42			F2 O/G	6723	Normal
43			F3 O/G	6724	Normal
44			F4 O/G	6725	Normal
45	WAREHOUSE	LDB-3	MAIN I/C	6726	Normal
46			R PHASE	6727	Normal
47			Y PHASE	6728	Normal
48			B PHASE	6729	Normal
49		LDB-2	MAIN I/C	6730	Normal
50			R PHASE	6731	Normal
51			Y PHASE	6732	Normal
52			B PHASE	6733	Normal
53		ELDB-1	MAIN I/C	6734	Normal
54	WAREHOUSE	D SCOPE ELDB-3	MAIN I/C	6735	Normal
55			R PHASE	6736	Normal
56			Y PHASE	6737	Normal
57			B PHASE	6738	Normal
58		D SCOPE LDB-3	MAIN I/C	6739	Normal
59			R PHASE	6740	Normal
60			Y PHASE	6741	Normal
61			B PHASE	6742	Normal
62		RPDB-1	R PHASE	6743	Normal
63			Y PHASE	6744	Normal
64			B PHASE	6745	Normal
65	WAREHOUSE	LDB	MAIN I/C	6746	Normal
66			R PHASE	6747	Normal
67			Y PHASE	6748	Normal
68			B PHASE	6749	Normal
69	SBD AREA WAREHOUSE	ELDB-3	MAIN I/C	6750	Alert
70		LDB-5	MAIN I/C	6751	Normal
71			R PHASE	6752	Normal
72			Y PHASE	6753	Normal
73			B PHASE	6754	Normal
74		MLDB AREA PANEL	MAIN I/C (INPUT)	6755	Normal
75			MAIN I/C (OUTPUT)	6756	Normal
76		O/G MCB (LDB1)		6757	Alert
77		O/G MCB (D SCOPE LDB3)		6759	Severe
78		O/G CABLE CHAMBER		6758	Normal

79		LDB-6	MAIN I/C	6760	Normal
80			R PHASE	6761	Normal
81			Y PHASE	6762	Normal
82			B PHASE	6763	Normal
83		LDB-9	MAIN I/C	6765	Normal
84			R PHASE	6766	Normal
85			Y PHASE	6767	Normal
86			B PHASE	6768	Normal
87	NEAR DOCK 24 WAREHOUSE SBD	LDB-02	MAIN I/C	6769	Normal
88	DOCK 24 AREA DSBD WAREHOUSE	ELDB-2	MAIN I/C	6770	Normal
89		DC DB	MAIN I/C	6771	Normal
90		EMDB-3	MAIN I/C	6772	Normal
91		LPDB-3	MAIN I/C	6773	Normal
92	DOCK 17 AREA	SDB-1 PANEL	MAIN I/C (CABLE TERMINATION)	6774	Normal
93			MAIN I/C (MCCB)	6775	Normal
94	DOCK AREA	SERVER UPS DB-2	DB	6776	Normal
95	DOCK B AREA	EMDB-2	MAIN MCB	6777	Normal
96			R PHASE	6778	Normal
97			Y PHASE	6779	Normal
98			B PHASE	6780	Normal
99	DOCK 7 AREA WAREHOUSE	LPDB-2	MAIN I/C	6781	Normal
100			R PHASE	6782	Normal
101			Y PHASE	6783	Normal
102			B PHASE	6784	Normal
103	DOCK 3 AREA WAREHOUSE	LDB1	MAIN I/C	6785	Normal
104		ELDB-1	MAIN I/C	6786	Normal
105		DCDB-1	MAIN I/C	6787	Normal
106	DOCK 7 AREA	HULUS FAN DB	MAIN BREAKER	6788	Normal
107	RAMP1 AREA WAREHOUSE	LDB-10	MAIN I/C	6789	Normal
108			R PHASE	6790	Normal
109			Y PHASE	6791	Normal
110			B PHASE	6792	Normal
111	TOILET AREA ENTRY GATE	RPDB-2	MAIN I/C	6794	Normal
112	UPS ROOM	MDB UPS PANEL	F1	6795	Normal
113			F2	6796	Normal

114			F3	6797	Normal
115	ELECTRICAL ROOM	MAIN ELDB	MAIN I/C	6798	Normal
116			O/G MCB-1	6799	Normal
117			O/G MCB-2	6800	Normal
118			O/G MCB-3	6801	Normal

Location	RACK SIDE
Measure Location	ELDB-4 MCB
Risk Category	Alert



Infrared Image



Normal Image

Description:

Image Number	6694
Target	Cable Termination

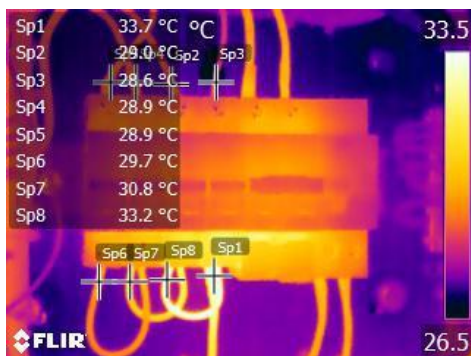
Data:

Measurement	Temp in Celsius
Max Temperature	35.2
Min Temperature	28.9
Delta T	6.3

Description and recommendation:

In the above thermo gram the temperature difference between Max and Min (ΔT) is 6.3°C. As per NFPA and NETA guidelines the above thermo gram is classified **Alert category**. It is recommended to check for Improper crimping, lug mismatch, or absence of lugs. Also recommended to check for tightness of the cable termination.

Location	SBD AREA WAREHOUSE
Measure Location	ELDB-3 MAIN I/C
Risk Category	Alert



Infrared Image



Normal Image

Description:

Image Number	6750
Target	Cable Termination

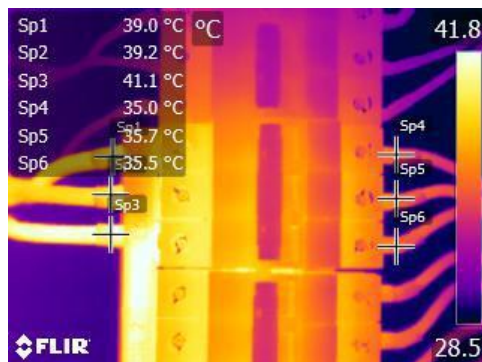
Data:

Measurement	Temp in Celsius
Max Temperature	33.6
Min Temperature	28.1
Delta T	5.1

Description and recommendation:

In the above thermo gram the temperature difference between Max and Min (ΔT) is 5.1°C. As per NFPA and NETA guidelines the above thermo gram is classified **Alert category**. It is recommended to check for Improper crimping, lug mismatch. Also recommended to check for tightness of the cable termination.

Location	SBD AREA WAREHOUSE
Measure Location	O/G MCB (LDB1)
Risk Category	Alert



Infrared Image



Normal Image

Description:

Image Number	6757
Target	Cable Termination

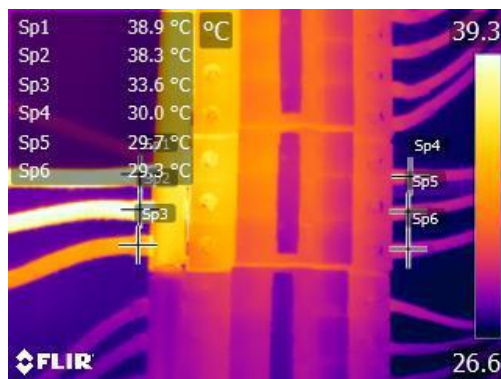
Data:

Measurement	Temp in Celsius
Max Temperature	41.1
Min Temperature	35.5
Delta T	5.6

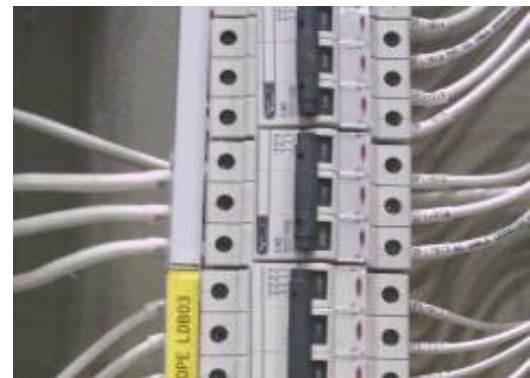
Description and recommendation:

In the above thermo gram the temperature difference between Max and Min (ΔT) is 5.6°C. As per NFPA and NETA guidelines the above thermo gram is classified **Alert category**. It is recommended to check for Improper crimping, lug mismatch, or absence of lugs. Also recommended to check for tightness of the cable termination.

Location	SBD AREA WAREHOUSE
Measure Location	O/G MCB (D SCOPE LDB3)
Risk Category	Severe



Infrared Image



Normal Image

Description:

Image Number	
Target	Cable Termination

Data:

Measurement	Temp in Celsius
Max Temperature	
Min Temperature	
Delta T	

Description and recommendation:

In the above thermo gram the temperature difference between Max and Min (ΔT) is 9.6°C. As per NFPA and NETA guidelines the above thermo gram is classified **Severe category**. It is recommended to check for Improper crimping, lug mismatch, or absence of lugs. Also recommended to check for tightness of the cable termination.

